

COURSE TITLE : CONTROL ENGINEERING
COURSE CODE : 4211
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER : 70
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
1	Basics of Control system and Laplace transformation	18
2	Transfer function ,Block Diagram and Signal Flow Graph	17
3	Time Response Analysis.	16
4	Stability Analysis.	19
TOTAL		70

COURSE OUTCOME

Module	G.O	On completion of the study of this Module the student will be able
1	1	To understand the basics of Control system
	2	To understand Laplace transformation and Inverse Laplace transformation
2	1	To Comprehend Transfer function
	2	To understand Block diagram
	3	To understand Signal Flow Graph
3	1	To understand Time Response Analysis of Control systems
	2	To realize Steady State Error and Error Constants
4	1	To understand Stability of control systems
	2	To understand Root Locus technique
	3	To understand Bode Plot .

On Completion of the study of this module the student will be able

MODULE I BASICS OF CONTROL SYSTEM AND LAPLACE TRANSFORMATION

1.1.0 To understand Basics of Control System

- 1.1.1 To define Control Systems
- 1.1.2 To explain the meaning of physical systems, physical model and mathematical model.
- 1.1.3 To describe linear time invariant and linear time variant system
- 1.1.4 To differentiate open loop and closed loop controls system, with physical examples

1.2.0 To understand Laplace transformation and Inverse Laplace transformation.

- 1.2.1 To define Laplace transforms.
- 1.2.2 To find Laplace transform of A, At, eat, e^{-at}, Sin at, Cos at, Multiplication by eat

- 1.2.3 To derive Real differentiation, Real integration, Initial value and Final value theorems.
- 1.2.4 To define Inverse Laplace transforms.
- 1.2.5 To find Inverse Laplace transformation by partial fraction method.
- 1.2.6 To solve simple problems (up to Third Order Systems)
- 1.2.7 To obtain the solution of simple linear differential equations

MODULE II TRANSFER FUNCTION, BLOCK DIAGRAM AND SIGNAL FLOW GRAPH

2.1.0 To Comprehend Transfer Functions of systems.

- 2.1.1 To define of transfer function and order of system.
- 2.1.2 To derive the transfer function of mechanical translational and rotational system.
- 2.1.3 To derive the transfer functions of RLC series and parallel circuit.
- 2.1.4 To describe the advantages and Disadvantages of transfer function.
- 2.1.5 To illustrate poles and Zeros of Transfer function.
- 2.1.6 To explain analogous systems.
- 2.1.7 To describe Force/Torque – voltage and Force/Torque– current analogy.

2.2.0 To understand Block Diagram algebra

- 2.2.1 To state the block diagram reduction rules.
- 2.2.2 To obtain the overall transfer function of Simple control systems by block diagram reduction rules (single input-single output systems).

2.3.0 To Understand signal flow graph.

- 2.3.1 To define the various parameters of the signal flow graph.
- 2.3.2 To state and explain Mason's gain formula.
- 2.3.3 To obtain the overall transmittance of control systems by signal flow graph (single input-single output systems).

MODULE III TIME RESPONSE ANALYSIS

3.1.0 To Understand Time response of control system

- 3.1.1 To explain transient response and Steady state response.
- 3.1.2 To describe standard test signals-step, ramp, parabolic & impulse signals
- 3.1.3 To obtain the time response of first order system for step, ramp and impulse inputs.
- 3.1.4 To define time constant.
- 3.1.5 To obtain the time response of second order system for step input (under damped).
- 3.1.6 To describe the transient response specifications of a system
- 3.1.7 To define characteristic equation, damping ratio, natural frequency and critical damping

3.2.0 To Realize Steady state error and Error Constants.

- 3.2.1 To define Type of a system
- 3.2.2 To explain Static error coefficients - Static position, velocity & acceleration error constants
- 3.2.3 To derive Steady state error in terms of K_p , K_v & K_a for Type 0, Type 1, Type 2 systems

MODULE IV STABILITY ANALYSIS

4.1.0 To Understand Stability of Control System

- 4.1.1 To define Absolute Stability and Relative stability
- 4.1.2 To explain Routh Hurwitz criterion
- 4.1.3 To solve simple problems using Routh Hurwitz criterion.

4.2.0 To Understand the Root Locus technique

- 4.2.1 To describe the Procedure to construct Root locus
- 4.2.2 To construct Root Locus for given transfer functions (Simple poles only)

4.3.0 To Understand Bode plot.

- 4.3.1 To define frequency response of a system
- 4.3.2 To explain Bode plot.
- 4.3.3 To draw Bode plot for simple transfer functions $.K, Ks, \frac{K}{s}, 1+Ts, 1-Ts, \frac{1}{1+Ts}, \frac{1}{1-Ts}$
- 4.3.4 To explain Frequency domain Specifications.

CONTENT DETAILS

MODULE I

Define Control Systems- meaning of physical systems, physical model and mathematical model- linear time invariant and linear time variant system- Differentiate open loop and closed loop controls system, with physical examples - Define Laplace transform- Laplace transform of A, At, eat, e^{-at}, Sin at, Cos at- Real differentiation, Real integration, Initial value and Final value theorems. Multiplication by eat-Inverse Laplace transforms- Inverse Laplace transformation by partial fraction method- simple problems (up to Third Order Systems) - solution of simple linear differential equations

MODULE II

Transfer function- order of system- Transfer function of mechanical translational- Mechanical rotational system RLC series circuit- RLC parallel circuit- Thermal system- Advantages and disadvantages of transfer function- Poles and zeros- Analogous systems force/Torque – Voltage and Force/Torque– Current analogy-Block diagram reduction rules-Obtain the overall transfer function of Simple control systems by block diagram reduction rules (single input-single output systems)- Various parameters of the signal flow graph- Mason's gain formula- Obtain the overall transmittance of control systems by signal flow graph (single input-single output systems).

MODULE III

Transient response- Steady state response -Standard test signals-Step, Ramp, Parabolic - Impulse signals - Time response of first order system-Step, Ramp , Impulse inputs-Time constant-Time response of second order system for step input-Transient response specifications of a system- Characteristic equation- Damping ratio- Natural frequency- Critical damping- Type of a system - Static

error coefficients - Static position, velocity ,acceleration error constants - Steady state error in terms of K_p, K_v, K_a for Type 0, Type 1, Type 2 systems

MODULE IV

Absolute Stability and Relative stability- Routh Hurwitz criterion Solve simple problems using Routh Hurwitz criterion- Procedure to construct Root locus-Construct Root Locus for given transfer functions (Simple poles only)- Frequency response of a system- Bode plot for simple transfer functions- $K, K_s, \frac{K}{s}, 1+Ts, 1-Ts, \frac{1}{1+Ts}, \frac{1}{1-Ts}$ - Frequency domain Specifications

REFERENCES

- 1 Nagoorkani , Control systems , RBA Publications
- 2 I J Nagrath.M. Gopal , Control Systems Engineering , New Age International
- 3 Katsuhiko Ogata , Modern Control Engineering , Pearson
- 4 B.S. Manke , Linear control systems , Khanna
- 5 S. Hasan Saeed -Automatic Control Systems , S. K. Kataria & Sons